



Continuum Mechanics

Steven M. Wise

swise1@utk.edu
University of Tennessee

Fall 2026



Chapter 03

Continuum Mass and Force

Part 1



Mass Density



Definition (Regular Region)

A region $B \subseteq \mathbb{E}^3$ is called *regular* if and only if

- ❶ B is the union of a finite number of open, disjoint, bounded components;
- ❷ the boundary surface ∂B is piecewise smooth and consists of a finite number of disjoint components;
- ❸ each component of ∂B is orientable, in the sense that it has two distinct sides.

Remark

A regular region may have a finite number of holes.



Definition (Mass Density)

Suppose that B is a regular region and that $\Omega \subseteq B$ is arbitrary. A smooth scalar field

$$\rho : B \longrightarrow \mathbb{R}$$

is called the *mass-density field* if

$$\rho(\mathbf{x}) > 0, \quad \mathbf{x} \in B.$$

The mass of the set Ω is

$$\text{mass}[\Omega] := \int_{\Omega} \rho(\mathbf{x}) \, dV_{\mathbf{x}}.$$

Volume and Units



The volume of Ω is denoted by

$$\text{vol}[\Omega] := \int_{\Omega} dV_x.$$

Typically, the units of the mass-density field are

$$[\rho] = \frac{\text{kg}}{\text{m}^3}.$$

Consequently,

$$[\text{mass}[\Omega]] = \text{kg}, \quad [\text{vol}[\Omega]] = \text{m}^3.$$



Remark (Recovering Density from Mass)

Suppose that $\Omega_\delta(\mathbf{x})$ is a family of open sets containing $\mathbf{x} \in B$ such that

$$\lim_{\delta \searrow 0} \text{vol}[\Omega_\delta(\mathbf{x})] = 0.$$

Then

$$\rho(\mathbf{x}) = \lim_{\delta \searrow 0} \frac{\text{mass}[\Omega_\delta(\mathbf{x})]}{\text{vol}[\Omega_\delta(\mathbf{x})]}.$$

In fact, one could define the mass density by assuming that this limit exists for every $\mathbf{x} \in B$.



Definition (Center of Mass)

Let B be a regular region with mass-density field $\rho : B \rightarrow \mathbb{R}$. The *center of mass* of $\Omega \subseteq B$ is

$$\mathbf{x}_{\text{com}}[\Omega] := \frac{1}{\text{mass}[\Omega]} \int_{\Omega} \mathbf{x} \rho(\mathbf{x}) \, dV_{\mathbf{x}}.$$

The factor $\rho(\mathbf{x})$ weights each point according to the mass near that point.



Definition (Center of Volume)

The *center of volume* of $\Omega \subseteq B$ is

$$\mathbf{x}_{\text{cov}}[\Omega] := \frac{1}{\text{vol}[\Omega]} \int_{\Omega} \mathbf{x} \, dV_{\mathbf{x}}.$$

It is possible that

$$\mathbf{x}_{\text{com}}[\Omega] \notin \Omega, \quad \mathbf{x}_{\text{cov}}[\Omega] \notin \Omega.$$

In general,

$$\mathbf{x}_{\text{com}}[\Omega] \neq \mathbf{x}_{\text{cov}}[\Omega].$$

However, if $\rho(\mathbf{x}) \equiv \rho_0$ on Ω , then

$$\mathbf{x}_{\text{com}}[\Omega] = \mathbf{x}_{\text{cov}}[\Omega].$$



Forces and Body Forces



Definition (Body Force)

A smooth vector field

$$\widehat{\mathbf{b}} : B \longrightarrow \mathcal{V}$$

is called a *body force* on B if it has units of force per unit volume:

$$[\widehat{\mathbf{b}}] = \frac{\text{force}}{\text{volume}} = \frac{\text{N}}{\text{m}^3}.$$

For $\Omega \subseteq B$, the resultant body force is

$$\mathbf{r}_b[\Omega] := \int_{\Omega} \widehat{\mathbf{b}}(\mathbf{x}) \, dV_{\mathbf{x}}, \quad [\mathbf{r}_b[\Omega]] = \text{N}.$$

Resultant Torque of a Body Force



The resultant torque on Ω about a fixed point $\mathbf{z} \in \mathbb{E}^3$ is

$$\boldsymbol{\tau}_b[\Omega, \mathbf{z}] := \int_{\Omega} (\mathbf{x} - \mathbf{z}) \times \widehat{\mathbf{b}}(\mathbf{x}) \, dV_{\mathbf{x}}.$$

The units of torque are

$$[\boldsymbol{\tau}_b[\Omega, \mathbf{z}]] = \text{N m}.$$

The cross product uses its linearity and its action on a right-handed orthonormal frame, as developed in Chapter 1.



Definition (Specific Body Force)

Let $\widehat{\mathbf{b}} : B \rightarrow \mathcal{V}$ be the body force per unit volume. The *body force per unit mass*, or *specific body force*, is

$$\mathbf{b}(\mathbf{x}) := \frac{\widehat{\mathbf{b}}(\mathbf{x})}{\rho(\mathbf{x})}.$$

Thus,

$$\mathbf{r}_b[\Omega] = \int_{\Omega} \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}},$$

and

$$\boldsymbol{\tau}_b[\Omega, \mathbf{z}] = \int_{\Omega} (\mathbf{x} - \mathbf{z}) \times \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}}.$$

Body Forces and Contact Forces



Body forces arise from external influences on a body, rather than from physical contact with another part of the body or with an external body.

For example, gravity is described by a specific body force $\mathbf{b}$, and the corresponding force per unit volume is $\rho\mathbf{b}$.

Surface forces, introduced next, describe forces induced by contact across a surface.



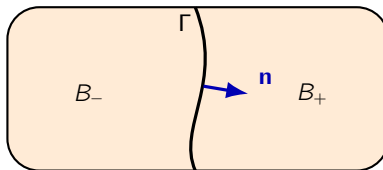
Surface Forces

An Internal Oriented Surface



Let Γ be an internal, imaginary boundary surface in a body B . The surface separates the body locally into a negative side B_- and a positive side B_+ .

Choose a unit normal field $\mathbf{n} : \Gamma \rightarrow \mathcal{V}$ that points away from B_- .



At each point $\mathbf{x} \in \Gamma$, the choice of $\mathbf{n}(\mathbf{x})$ determines the positive and negative sides of the surface.



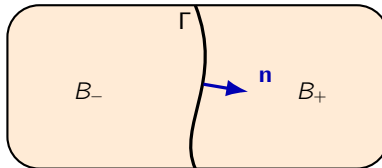
Definition (Traction on an Internal Surface)

The force per unit area exerted by material on the positive side upon material on the negative side is represented by a function

$$\mathbf{t}_n : \Gamma \longrightarrow \mathcal{V}.$$

The vector field $\mathbf{t}_n$ is called the *traction*, or *surface-force field*, for Γ .

The subscript records the chosen orientation. Reversing the normal reverses which side exerts force on which material.



Traction on the Boundary of the Body



When $\Gamma = \partial B$, choose $\mathbf{n}$ as the outward-pointing unit normal. Then

$$\mathbf{t}_n : \partial B \longrightarrow \mathcal{V}$$

is the force per unit area exerted by the external body on the body B .

The units of the traction vector are

$$[\mathbf{t}_n] = \frac{\text{N}}{\text{m}^2}.$$



Definition (Resultants of a Surface Traction)

Let Γ be an oriented surface in B , and let $\mathbf{t}_n : \Gamma \rightarrow \mathcal{V}$ be the surface traction. Its resultant force is

$$\mathbf{r}_s[\Gamma] := \int_{\Gamma} \mathbf{t}_n(\mathbf{x}) \, dA_x.$$

The resultant torque about $\mathbf{z} \in \mathbb{E}^3$ is

$$\boldsymbol{\tau}_s[\Gamma, \mathbf{z}] := \int_{\Gamma} (\mathbf{x} - \mathbf{z}) \times \mathbf{t}_n(\mathbf{x}) \, dA_x.$$

Thus,

$$[\mathbf{r}_s[\Gamma]] = N, \quad [\boldsymbol{\tau}_s[\Gamma, \mathbf{z}]] = N \, \text{m}.$$



Cauchy's Postulate



Axiom 3.1 (Cauchy's Postulate)

The traction field $\mathbf{t}_n$ on a surface Γ in B depends only pointwise on the unit normal field

$$\mathbf{n} : \Gamma \longrightarrow \mathcal{V}.$$

In particular, there is a function

$$\mathbf{t} : \mathcal{N} \times B \longrightarrow \mathcal{V},$$

where $\mathcal{N} \subset \mathcal{V}$ is the set of unit vectors, such that

$$\mathbf{t}_n(\mathbf{x}) = \mathbf{t}(\mathbf{n}(\mathbf{x}), \mathbf{x}).$$

The function $\mathbf{t} : \mathcal{N} \times B \rightarrow \mathcal{V}$ is called the *traction function* for B .



Result 3.2

Let $\mathbf{t} : \mathcal{N} \times B \rightarrow \mathcal{V}$ be the traction function for a body B . Suppose that $\mathbf{t}(\mathbf{n}, \mathbf{x})$ is continuous and that, for every sequence of regular regions $\Omega \subseteq B$ whose volumes tend to zero,

$$\frac{1}{\text{Area}(\partial\Omega)} \int_{\partial\Omega} \mathbf{t}(\mathbf{n}(\mathbf{x}), \mathbf{x}) \, dA_{\mathbf{x}} \longrightarrow \mathbf{0} \quad \text{as } \text{vol}[\Omega] \longrightarrow 0. \quad (3.1)$$

Then

$$\mathbf{t}(-\mathbf{n}, \mathbf{x}) = -\mathbf{t}(\mathbf{n}, \mathbf{x}), \quad \mathbf{n} \in \mathcal{N}, \quad \mathbf{x} \in B.$$

Proof.

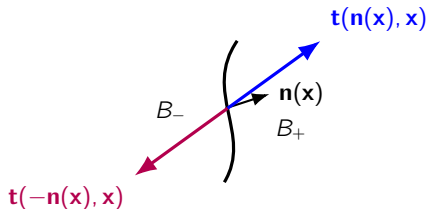
See Gonzalez and Stuart, page 81.





Remark

The traction exerted by material on the positive side upon material on the negative side is equal and opposite to the traction exerted by the material on the negative side upon the material on the positive side.





The Stress Tensor



Result 3.3 (Cauchy's Theorem)

Let $\mathbf{t} : \mathcal{N} \times B \rightarrow \mathcal{V}$ be the traction function for a body B , and suppose that it satisfies the conditions of Result 3.2. Then $\mathbf{t}(\mathbf{n}, \mathbf{x})$ is linear in $\mathbf{n}$. More precisely, for each $\mathbf{x} \in B$ there is a second-order tensor $S(\mathbf{x}) \in \mathcal{V}^2$ such that

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) = S(\mathbf{x})\mathbf{n}.$$

The field

$$S : B \longrightarrow \mathcal{V}^2$$

is called the *Cauchy stress field* for B .

Proof.

See Gonzalez and Stuart, pages 83–84.



Components of the Cauchy Stress



Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame. The components of the stress tensor are

$$S_{ij}(\mathbf{x}) = \mathbf{e}_i \cdot (S(\mathbf{x})\mathbf{e}_j).$$

Thus,

$$S(\mathbf{x}) = S_{ij}(\mathbf{x})\mathbf{e}_i \otimes \mathbf{e}_j.$$

If

$$\mathbf{t}(\mathbf{n}(\mathbf{x}), \mathbf{x}) = t_i(\mathbf{n}(\mathbf{x}), \mathbf{x})\mathbf{e}_i, \quad \mathbf{n}(\mathbf{x}) = n_j(\mathbf{x})\mathbf{e}_j,$$

then

$$t_i(\mathbf{n}(\mathbf{x}), \mathbf{x}) = S_{ij}(\mathbf{x})n_j(\mathbf{x}).$$

Here i is free and j is dummy.

Stress Components as Coordinate-Plane Traction



Take the normal to be the frame vector $\mathbf{e}_k$. Then

$$\begin{aligned} t_i(\mathbf{e}_k, \mathbf{x}) &= S_{ij}(\mathbf{x})[\mathbf{e}_k]_j \\ &= S_{ij}(\mathbf{x})\delta_{jk} \\ &= S_{ik}(\mathbf{x}). \end{aligned}$$

Therefore,

$$S_{ij}(\mathbf{x}) = t_i(\mathbf{e}_j, \mathbf{x}).$$

The j th column of the matrix of S contains the components of the traction on the coordinate plane whose unit normal is $\mathbf{e}_j$.

Tractions on the Coordinate Planes

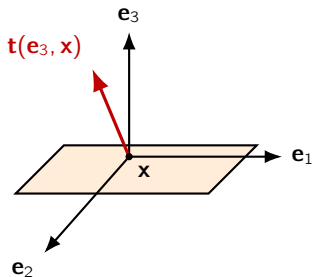


The three column vectors of the matrix of S are

$$\mathbf{t}(\mathbf{e}_1, \mathbf{x}) = S_{i1}(\mathbf{x})\mathbf{e}_i,$$

$$\mathbf{t}(\mathbf{e}_2, \mathbf{x}) = S_{i2}(\mathbf{x})\mathbf{e}_i,$$

$$\mathbf{t}(\mathbf{e}_3, \mathbf{x}) = S_{i3}(\mathbf{x})\mathbf{e}_i.$$





Equilibrium

Loading of a Body



Assume that a body B has mass density

$$\rho : B \longrightarrow \mathbb{R},$$

applied external traction

$$\bar{\mathbf{t}} : \partial B \longrightarrow \mathcal{V},$$

and applied body force per unit mass

$$\mathbf{b} : B \longrightarrow \mathcal{V}.$$

The fields ρ , $\bar{\mathbf{t}}$, and $\mathbf{b}$ are assumed to be time independent. We want to determine when B is in mechanical equilibrium.



Axiom 3.4 (Conditions for Equilibrium)

If a body in configuration B is in mechanical equilibrium, then the resultant force and the resultant torque about any fixed point $\mathbf{z}$ vanish for every open subset $\Omega \subseteq B$.

Writing $\mathbf{t}(\mathbf{x}) = \mathbf{t}(\mathbf{n}(\mathbf{x}), \mathbf{x})$ on $\partial\Omega$, the conditions are

$$\begin{aligned}\mathbf{r}[\Omega] &:= \int_{\Omega} \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{t}(\mathbf{x}) \, dA_{\mathbf{x}} = \mathbf{0}, \\ \boldsymbol{\tau}[\Omega, \mathbf{z}] &:= \int_{\Omega} (\mathbf{x} - \mathbf{z}) \times \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial\Omega} (\mathbf{x} - \mathbf{z}) \times \mathbf{t}(\mathbf{x}) \, dA_{\mathbf{x}} = \mathbf{0}.\end{aligned}\tag{3.7}$$



Proposition

Suppose that $\mathbf{r}[\Omega] = \mathbf{0}$ for every open $\Omega \subseteq B$. Then, for every fixed point $\mathbf{z}$,

$$\begin{aligned} \int_{\Omega} (\mathbf{x} - \mathbf{z}) \times \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial\Omega} (\mathbf{x} - \mathbf{z}) \times \mathbf{t}(\mathbf{x}) \, dA_{\mathbf{x}} \\ = \int_{\Omega} \mathbf{x} \times \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{x} \times \mathbf{t}(\mathbf{x}) \, dA_{\mathbf{x}}. \end{aligned}$$

Proof.

Using linearity of the cross product, the difference between the two sides is

$$\begin{aligned} -\mathbf{z} \times \int_{\Omega} \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} - \mathbf{z} \times \int_{\partial\Omega} \mathbf{t}(\mathbf{x}) \, dA_{\mathbf{x}} \\ = -\mathbf{z} \times \mathbf{r}[\Omega] = \mathbf{0}. \end{aligned}$$





Thus, once resultant force vanishes, equation (3.7) may be replaced by the equivalent pair

$$\begin{aligned}\mathbf{r}[\Omega] &:= \int_{\Omega} \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{t}(\mathbf{x}) \, dA_{\mathbf{x}} = \mathbf{0}, \\ \boldsymbol{\tau}[\Omega, \mathbf{0}] &:= \int_{\Omega} \mathbf{x} \times \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{x} \times \mathbf{t}(\mathbf{x}) \, dA_{\mathbf{x}} = \mathbf{0}.\end{aligned}\tag{3.7}$$



Local Equilibrium Conditions



Result 3.5 (Local Equilibrium Conditions)

Assume that the Cauchy stress field S is continuously differentiable and that the density field ρ and specific body-force field $\mathbf{b}$ are continuous. Then the equilibrium conditions in equation (3.7) are equivalent to

$$\begin{aligned}\nabla \cdot S(\mathbf{x}) + \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) &= \mathbf{0}, & \mathbf{x} \in B, \\ S^T(\mathbf{x}) &= S(\mathbf{x}), & \mathbf{x} \in B.\end{aligned}\tag{3.8}$$



Proof.

Assume first that the equilibrium conditions in equation (3.7) hold. By Cauchy's theorem,

$$\mathbf{t}(\mathbf{x}) = \mathbf{S}(\mathbf{x})\mathbf{n}(\mathbf{x}) \quad \text{on } \partial\Omega.$$

The force balance therefore gives

$$\int_{\Omega} \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{S}(\mathbf{x})\mathbf{n}(\mathbf{x}) \, dA_{\mathbf{x}} = \mathbf{0}.$$

Using the tensor divergence theorem,

$$\int_{\partial\Omega} \mathbf{S}\mathbf{n} \, dA = \int_{\Omega} \nabla \cdot \mathbf{S} \, dV.$$



Proof (Cont.)

For every open $\Omega \subseteq B$,

$$\int_{\Omega} (\nabla \cdot \mathbf{S} + \rho \mathbf{b}) \, dV = \mathbf{0}.$$

The integrand is continuous. Applying the localization theorem component by component gives

$$\nabla \cdot \mathbf{S} + \rho \mathbf{b} = \mathbf{0} \quad \text{in } B.$$

This proves the first local equilibrium equation in (3.8).



Proof (Cont.)

The torque balance, again using $\mathbf{t} = S\mathbf{n}$, is

$$\int_{\partial\Omega} \mathbf{x} \times (S\mathbf{n}) \, dA + \int_{\Omega} \mathbf{x} \times \rho \mathbf{b} \, dV = \mathbf{0}.$$

Since the force equation gives

$$\rho \mathbf{b} = -\nabla \cdot S,$$

we obtain

$$\int_{\partial\Omega} \mathbf{x} \times (S\mathbf{n}) \, dA - \int_{\Omega} \mathbf{x} \times (\nabla \cdot S) \, dV = \mathbf{0}. \quad (3.9)$$



Proof (Cont.)

Choose a right-handed orthonormal frame and define the second-order tensor R by

$$R_{i\ell} := \epsilon_{ijk} x_j S_{k\ell}, \quad R = R_{i\ell} \mathbf{e}_i \otimes \mathbf{e}_\ell.$$

Then

$$\begin{aligned} [\mathbf{R}\mathbf{n}]_i &= R_{i\ell} n_\ell \\ &= \epsilon_{ijk} x_j S_{k\ell} n_\ell \\ &= [\mathbf{x} \times (\mathbf{S}\mathbf{n})]_i. \end{aligned}$$

Hence

$$\mathbf{R}\mathbf{n} = \mathbf{x} \times (\mathbf{S}\mathbf{n}).$$



Proof (Cont.)

Equation (3.9) becomes

$$\int_{\partial\Omega} R\mathbf{n} \, dA - \int_{\Omega} \mathbf{x} \times (\nabla \cdot S) \, dV = \mathbf{0}. \quad (3.10)$$

The tensor divergence theorem implies

$$\int_{\Omega} (\nabla \cdot R - \mathbf{x} \times (\nabla \cdot S)) \, dV = \mathbf{0}.$$

Since $\Omega \subseteq B$ is arbitrary, localization gives

$$\nabla \cdot R - \mathbf{x} \times (\nabla \cdot S) = \mathbf{0} \quad \text{in } B.$$



Proof (Cont.)

In component form, the preceding identity is

$$(\epsilon_{ijk} x_j S_{kl})_{,l} - \epsilon_{ijk} x_j S_{kl,l} = 0.$$

The first term may be expanded as

$$\begin{aligned} (\epsilon_{ijk} x_j S_{kl})_{,l} &= \frac{\partial}{\partial x_l} (\epsilon_{ijk} x_j S_{kl}) \\ &= \epsilon_{ijk} \delta_{jl} S_{kl} + \epsilon_{ijk} x_j S_{kl,l} \\ &= \epsilon_{ijk} S_{kj} + \epsilon_{ijk} x_j S_{kl,l}. \end{aligned}$$

Therefore the two terms containing $x_j S_{kl,l}$ cancel, and

$$0 = \epsilon_{ijk} S_{kj}.$$

Here i is free, while j, k are dummy indices.



Proof (Cont.)

We now expand the contraction for each value of the free index i . For $i = 1$,

$$\begin{aligned} 0 &= \epsilon_{1jk} S_{kj} \\ &= \epsilon_{123} S_{32} + \epsilon_{132} S_{23} \\ &= S_{32} - S_{23}, \end{aligned}$$

so $S_{32} = S_{23}$.

For $i = 2$,

$$\begin{aligned} 0 &= \epsilon_{2jk} S_{kj} \\ &= \epsilon_{231} S_{13} + \epsilon_{213} S_{31} \\ &= S_{13} - S_{31}, \end{aligned}$$

so $S_{13} = S_{31}$.



Proof (Cont.)

Finally, for $i = 3$,

$$\begin{aligned} 0 &= \epsilon_{3jk} S_{kj} \\ &= \epsilon_{312} S_{21} + \epsilon_{321} S_{12} \\ &= S_{21} - S_{12}, \end{aligned}$$

so $S_{21} = S_{12}$. Together with the diagonal identities, these three cases show that

$$S_{ij} = S_{ji}, \quad i, j \in \{1, 2, 3\}.$$

Hence $S^T = S$.

Conversely, assume the two local equations in (3.8). Integrating them over an arbitrary $\Omega \subseteq B$ and applying the tensor divergence theorem recovers the force and torque balances in (3.7). The details are left as an exercise. $\square$